

Design Thinking, Innovation and Entrepreneurship (25HU102)

L T P 2 0 0

Credit 02

Lectures 30

Objective: The objective of this course is to familiarize students with the design thinking process as a tool for problem-solving. It aims to equip students with design thinking skills and ignite their minds to create innovative ideas, develop solutions for real-time problems

Course Outcome: After completion of the course, the students will be able to:

CO1	Differentiate conventional problem-solving from design thinking through case studies and team dynamics.	K2
CO2	Apply the stages of the design thinking framework to develop user-centered solutions.	K3
CO3	Apply empathy, ideation, prototyping, and testing techniques to generate and refine solutions.	K3
CO4	Analyse types of innovation, assess value creation, and evaluate readiness using TRL.	K4 and K5
CO5	Apply IPR concepts and entrepreneurship tools to design a business model and pitch an ideas.	K5

Unit 1: Design Thinking Introduction

Introduction to Design Thinking, conventional problem solving versus design thinking, business uses of design thinking, variety within the design thinking discipline, formation and importance of the design thinking team, workshops and meetings.

Case Study: Mumbai Dabbawallas, Tata Nano, Jeevan Bindi

Unit 2: Design Thinking Framework

Design thinking approaches in stages: empathize with user, Moccasin Walk, define the problem, Ideate, Prototype (alternate solution), test the solution. Problem identification techniques and research methods

Case Study: Portable Ultrasound Machine (GE V scan), Embrace Infant Warmer, Tally, Lucky Iron Fish

Activity: Presentation (Case study)

Unit 3: Design thinking techniques

Listening and Empathising techniques-engagement, observation, interviews, surveys, Define-Create problem statements, identify user needs and pain points, synthesize research data, Ideate- Pattern recognition, scamper techniques, and connecting the dots, Prototype- Paper prototypes, Wireframes, Models or role-play scenarios, Forms of testing in design thinking

Case study: Pure it, Royal Enfield, Big Basket, IDEO Water Pump, Airbnb, Oral-B toothbrush

Activity: Empathy map, Journey map

Unit 4: Introduction to innovation

Concept of innovation, innovation and creativity, difference between invention and innovation, types of innovation, product vs process innovation, value creation, social and grassroots innovation, benefits of innovation, Technology Readiness Level (measuring innovation),

Unit 5: IPR and Entrepreneurship

Introduction to IPR, Types of IPR (copyright, patent, trademark, GI tag), advantages and disadvantages of IPR, prior art research techniques using tools such as Google Patent and other

Entrepreneurship concept, myths versus reality of entrepreneurship, entrepreneurship competencies, sensing business opportunity, Business model canvas and value proposition, start-up success and failure story, idea pitching (pitch deck development)

Text Books

1. The Design of Everyday Things by Don Norman. Publisher: Basic Books (Revised and Expanded Edition, 2013)
2. Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving by Pavan Soni. Publisher: Penguin Random House India (2020)
3. Design Thinking: A Beginner's Perspective by E. Balagurusamy. Publisher: McGraw Hill Education India (2024)

Reference Books

1. Change by Design: How Design Thinking Creates New Alternatives for Business and Society by Tim Brown, Publisher: Harper Business (2009)
2. Creative Confidence: Unleashing the Creative Potential Within Us All by Tom Kelley & David Kelley. Publisher: Crown Business (2013)
3. Design Thinking: A Framework for Applying Design Thinking in Problem Solving by Anuja Agarwal. Publisher: Cengage India (2024)